

Lexical Conceptual Mapping

Connecting ChainNet and WordNet: Metaphor Detection

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Core Objectives

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- Testing the automated connection between **ChainNet** and **WordNet**.
- Investigating conceptual metaphor schemas:
 - COLOUR IS MINERAL (Ruby, Emerald...)
 - COLOUR IS PLANT (Rose, Violet...)
- Building a processing pipeline to explore semantic hierarchies.

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Climbing the Semantic Tree

- The algorithm traverses **hypernyms** (broader categories) up to **10 levels**.
- **Example:** Emerald → Beryl → Mineral.
- **Blacklist:** Filtering generic words like *entity*, *color*, *flora* to reduce semantic noise and ensure precise mapping.

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The "Gold-God" Mapping Issue

- **Observation:** Initial runs showed unexpected links between yellow shades and theology.
- **Cause:** A lexical mismatch during dictionary mapping (the word *"gold"* was erroneously processed as *"god"*).
- **Consequence:** The WordNet tree correctly followed the path for *"deity"*, which highlighted the extreme sensitivity of the mapping logic to small typos.
- **Correction:** Strict verification of the mapping dictionary was implemented to maintain domain consistency.

Lesson: A single character difference ('l') shifted the entire domain from Geology to Theology.

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Case Study: Data Consistency (Aquamarine)

While investigating the absence of certain terms like *aquamarine*, we applied three levels of verification:

- 1 **Internal String Scan:** Systematic check for the substring in the source JSON.
- 2 **Hypernym Path Analysis:** Testing if WordNet ancestors would trigger a mapping if the term were present.
- 3 **Lemma Inventory:** Generating a list of all available lemmas in the dataset.

Result: The term was confirmed to be physically absent from the source dataset, validating that the mapping gaps were data-related, not code-related.

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Technical Verification: Python Logic

1. Searching for "aquamarine" in JSON

```
1 # Check if the lemma exists in the raw data list
2 all_lemmas = [entry["lemma"] for entry in chainnet_data]
3 print(f"Is 'aquamarine' present? {'aquamarine' in all_lemmas}")
```

2. WordNet Traversal Test (Colour vs Mineral)

```
1 # Trying to force a path through ancestors
2 ancestors = get_hypernyms(synset_id)
3 if "mineral" in ancestors or "color" in ancestors:
4     # Logic worked for 'ruby' but found nothing for 'aquamarine'
5     pass
```

3. Dataset Inventory Print

```
1 # Listing unique words to see what the dataset actually contains
2 unique_lemmas = sorted(list(set(all_lemmas)))
3 print(unique_lemmas[:10])
```



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Mapping Results: Dual Nature of Concepts

Domain	Examples	Physical Object	Color Mapping
MINERAL	Ruby	Precious gem	→ Deep red
MINERAL	Emerald	Green beryl	→ Vivid green
MINERAL	Coral	Marine skeleton	→ Pinkish-orange
PLANT	Rose, Violet	Flower/Shrub	→ Pink/Purple

- **Observation:** All identified lemmas exhibit a dual semantic role, acting as both a **source domain** (object) and a **target domain** (color).
- Mapping refined by implementing semantic blacklists.
- Output generated in a structured format for further review.

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Summary and Future Outlook

- **Current Status:** Established a functional pipeline that successfully maps core concepts from the provided data.
- **Methodology:** Semantic blacklisting effectively narrowed results to relevant domains, avoiding incorrect links (e.g., *indigo*).
- **Next Steps:** The system is designed to be scalable for future dataset expansions (e.g., adding *amethyst* or *orchid*).

Thanks for your attention!

- We welcome your feedback and questions.